XUEJUN (JACK) CHENG

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RESEARCH EXPERIENCE

Research Project (first author), with Rena Wei Gao, LLM Foundation Team, ByteDance / University of Melbourne (under review, ACL Rolling Review, Oct 2026)

- Examined whether common annotation quality metrics (inter-annotator agreement, classifier F1, LLM self-consistency and LLM replacement tests) actually establish construct validity in multilingual NLP
- Evaluated 1,272 multilingual comments across seven theory-derived constructs and identified three failure mechanisms: bundled definitions, extreme label sparsity and collapse onto surface variables
- Benchmarked TF-IDF, XLM-R and mDeBERTa under grouped, leakage-safe evaluation, and compared internal metrics with external evidence from an independent 550-participant survey

Independent Research Project, Emotion-Aware Narrative World Model (ongoing)

- Developing a narrative world model for long-horizon decision-making in *Detroit: Become Human*, covering emotional state, relationships, morality, story progression and narrative arc
- Exploring multi-objective planning that balances successful outcomes with emotionally and narratively meaningful choices, rather than optimising for a single "best ending"
- Designing evaluation around affect prediction, relationship dynamics, moral decisions and long-horizon planning

Research Project, Monash University, with Prof. Wong Wai Peng (under review at ICWSM 2027)

- Built a cross-platform corpus of 38,271 comments on seven girl groups, collected from seven social media platforms, with multilingual text, linked media, platform and geographic information
- Designed a leakage-safe multimodal evaluation setup using video-disjoint splits, wrong-video controls, media-only probes, CLIP embeddings and mixed-effects models
- Found that aggregated media features gave no measurable predictive gain once video-identity leakage was controlled, while platform (venue) remained significantly associated with cultural framing

Research Project, Star IP and Cultural Export (presented at GAAC 2026)

- Developed a mixed-method framework combining PLS-SEM with multilingual social media comment analysis to study star IP, tourism promotion, M-Pop acceptance and cultural export
- Analysed 550 valid survey responses across eight latent constructs, testing 12 structural paths with reliability and validity checks, bootstrapping, MICOM and multi-group analysis
- Built and annotated a corpus of 1,500 comments from 21 content groups; implemented TF-IDF + logistic regression, XLM-R with label-wise attention and asymmetric loss, and validation-tuned late fusion

Literature Review, Zhejiang University–UIUC Institute (Jul 2025)

- Structured literature review of major computer vision research directions, representative models and recent methods, comparing model design, application scenarios and limitations (grade: A)

RELEVANT RESEARCH SKILLS

- Programming in Python; building and training neural networks from scratch in PyTorch (hand-implemented forward and backward passes, CNNs); scikit-learn
- LLM agents e.g. designing multi-step AI agent workflows for an HR evaluation system (code on GitHub)
- Transformer models e.g. fine-tuning and modifying XLM-R and mDeBERTa with label-wise attention, asymmetric loss for label imbalance and late fusion of multiple models
- Multimodal learning e.g. CLIP feature extraction for video-linked social media data
- Experimental design e.g. controlled comparisons, leakage-safe grouped and video-disjoint splits, mismatched-video controls, attention to reproducibility
- Statistical analysis e.g. mixed-effects models, PLS-SEM (bootstrapping, MICOM, multi-group analysis)
- Search and planning algorithms e.g. heuristic search, adversarial search

EDUCATION

Monash University (Feb 2026 – Feb 2028, expected)

Master of Artificial Intelligence, GPA 3.25/4

Units include Deep Learning, Planning and Automated Reasoning, Fundamentals of AI, Python Programming

Georgia Institute of Technology (Feb 2027 – Feb 2029, expected)

MS Computer Science

Planned coursework: Machine Learning, Reinforcement Learning, AI for Robotics

Shanghai Jiao Tong University, KoGuan School of Law (Feb–Dec 2025)
Exchange Program, Grade A
Contract law, financial market regulation; research paper on the legal regulation of AI applications in financial markets

Coventry University (2021–2025)
BA (Hons) Marketing 2:1

ADDITIONAL RELEVANT EXPERIENCE

NUS–MSRA AI Summer School (Aug 2026)
Participant in the AI summer school jointly organised by the National University of Singapore and Microsoft Research Asia

SLAI AI + Quantum Winter School, Shenzhen Loop Area Institute (2025)
Reproduced an LLM-based quantum circuit simulation

Student Contributor, AI Academic Reviewer Skill Project, Prof. Liu's team (2026)
Collecting and logging public materials on selected Management Science / Marketing Science scholars, with reliability grading, as source data for an LLM reviewing skill

Course Representative, Faculty of IT, Monash University (Feb 2026–present)
Course representative and member of the student support team

Course Representative, Coventry University Students' Union (2021–2025)
Represented students on the course; also a member of the Phoenix Society and the Economics Society

ADDITIONAL SKILLS & COURSES

- Coursework projects: Pac-Man search and adversarial search agents (Monash, 2026); feedforward network from scratch on MNIST, FashionMNIST tuning, CNNs and a Kaggle competition (Monash Deep Learning, 2026)
- Conference presentations: UNU Macau AI Conference (2025); GAAC 2026
- Languages: Mandarin (native); English (fluent, degree-level study in the UK and Australia)

PUBLICATIONS

2026 Optimizing the logistics and warehousing model of anti-cancer drugs for cancer patients. *Psycho-Oncologie*. doi:10.18282/po5579 (corresponding author)

2025 Simulation study on optimization of delivery path for community group buying. *Frontiers in Future Transportation* 6. doi:10.3389/ffutr.2025.1671246 (corresponding author)

2025 Cheng X. Future legal regulation of AI agent applications in financial markets. SSRN, 1 July 2025. doi:10.2139/ssrn.6631898 (sole author; presented at UNU Macau AI Conference 2025)

MANUSCRIPTS UNDER REVIEW

Cheng X., Xu X., Gao R. W. High agreement, wrong construct: what annotation quality metrics actually certify. Under review, ACL Rolling Review (October 2026 cycle)

Screen and speech: aggregated media features do not predict fandom discourse, but venue covaries with it. Under review, ICWSM 2027 (with Prof. Wong Wai Peng)

CONFERENCE PAPERS

From star IP to cultural export: integrating PLS-SEM and construct-aligned multilingual comment analytics. GBA AI Alliance Conference (GAAC 2026), "AI Innovation and Applications: Shaping the Intelligent Economy", 12 September 2026. DOI pending